

MERCH STORE ANALYTICS REVIEW

Where the Google Merchandise Store is losing revenue, which channels really drive sales, and what to fix first

An analysis of the GA4 public sample, 1 November 2020 to 31 January 2021, with the historical data corrected for four tracking faults.

**\$338.3K**

Revenue, deduplicated

**5,288**

Orders

**360.1K**

Sessions

**1.47%**

Conversion rate

THE SHORT VERSION



Fix the tracking, protect search, and work on the product page



Tracking errors overstated revenue by 7% and credited 30% of orders to the store's own website.

→ [Fix GA4 tracking and add alerts for when tags stop firing](#)



Organic Search is credited with 40 to 43% of sales under all seven attribution models.

→ [Protect SEO; judge Paid Search on more than last-click numbers](#)



3 in 4 product viewers never add to basket: the biggest leak.

→ [A/B test the product page, then checkout](#)



January earned 57% less per day than December because conversion fell, not traffic.

→ [Win the second purchase within two weeks of the first](#)

MEASUREMENT

 1. Measurement reliability

4

tracking problems would have misled every report built on the raw data.



7%

Revenue overstated by
duplicate purchase events

837

Real orders with no
transaction ID, worth \$30.7K,
that standard deduplication
throws away

22 days

Add-to-cart tag broken in
November, while people kept
checking out

6 days

Order values missing from
26 to 31 January, about
\$12K to \$16K of revenue**BUILT IN** Each problem is detected and flagged automatically in the pipeline, with data tests.

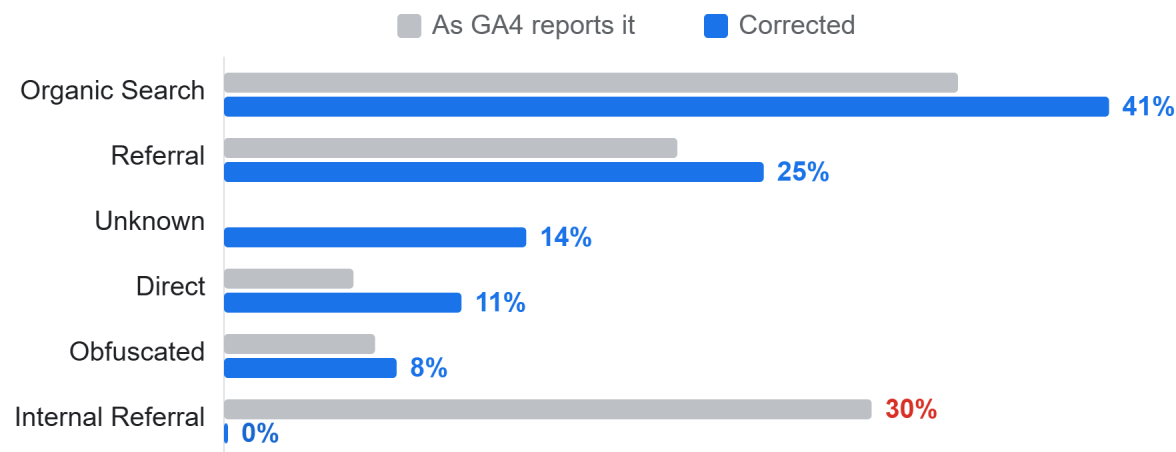
MEASUREMENT

Self-referrals

30% of orders were credited to the store's own website.

Self-referrals: 870 orders go back to real channels, 719 have no known source

Share of last-click orders by channel, before and after the corrections



1,589 orders were last-click credited to the store itself: customers coming back from the payment page start a new session.

Removing them hands 870 orders back to real channels (Organic Search +19%, Referral +17%). The other 719 had no earlier visit, so their source is Unknown.

→ NEXT ACTION
Add the store and payment domains to GA4's unwanted referrals list.

CHANNELS

2. Which channels get credit for sales

40–43% Organic Search is credited with 40 to 43% of sales under all seven attribution models.

Organic Search gets 40 to 43% of conversions under every model

Share of conversions credited to each channel, by model · darker means more credit

	Last click	Last non-direct	First click	Linear	Time decay	Position-based	Data-driven
Organic Search	40.9%	43.0%	42.6%	41.4%	41.6%	41.6%	40.0%
Referral	24.9%	25.8%	24.2%	24.2%	24.6%	24.4%	23.3%
Direct	10.7%	6.8%	11.1%	11.4%	11.0%	11.1%	13.2%
Unknown	13.6%	13.6%	13.6%	13.6%	13.6%	13.6%	12.2%
Obfuscated	8.2%	9.0%	6.6%	7.6%	7.5%	7.5%	9.2%
Paid Search	1.2%	1.3%	1.5%	1.3%	1.3%	1.3%	1.5%
Email	0.5%	0.5%	0.4%	0.5%	0.5%	0.5%	0.5%
Affiliates	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.2%

The result doesn't depend on the method: all seven models sit within **3 points**.

Paid Search gets 23% more credit from the data-driven model than from last click: it starts journeys other channels finish.

INTERPRETATION
65% of orders come from a single visit, where every model agrees.

CONVERSION

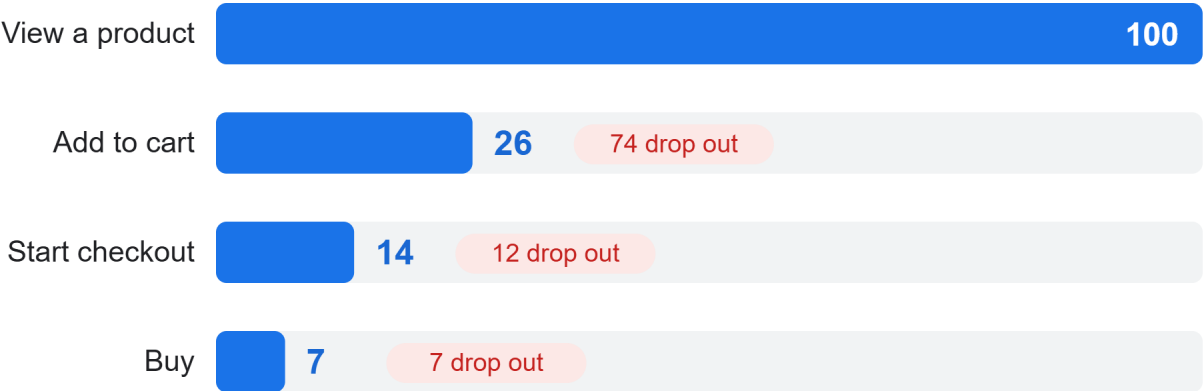
 3. Where the journey loses buyers

74/100

product viewers never add to basket, the biggest leak in the funnel.

Of 100 people who view a product, 74 leave before the basket

What happens to 100 product viewers · days with working cart tracking



Of 100 product viewers, 26 add to cart (25.7%), 14 start checkout and 7 buy.

Only **1.35%** of visits end in a purchase.

After the basket, about half of carts never reach checkout, and half of checkouts never finish.

→ **NEXT ACTION**

Test delivery information, reviews and images on product pages, then checkout.
No clear difference between mobile and desktop.

CONVERSION

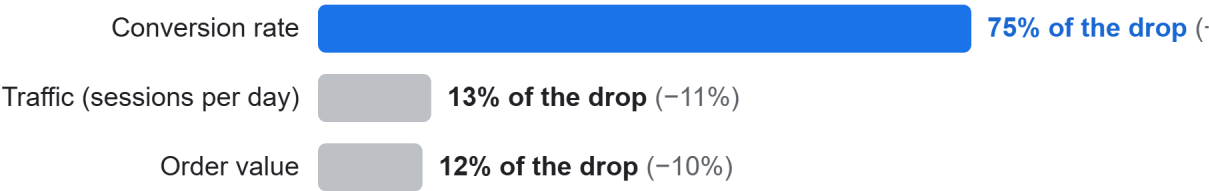
 4. Why January fell

57%

less revenue per day in January than December, and conversion explains it.

Conversion explains three quarters of January's 57% drop

Share of the December-to-January drop in revenue per day



	DECEMBER	JANUARY	CHANGE
Conversion rate	1.79%	0.98%	-0.81 pts
Sessions	133,351	118,377	-11%
Revenue per day*	\$5,101	\$2,174	-57%

*Days with recorded order values (26 to 31 January excluded).

Visits fell only 11% from December to January.

Conversion fell from 1.79% to 0.98%: people still came, they stopped buying.

→ NEXT ACTION

Plan Q1 on a post-holiday baseline, and test January offers on the product page.

CUSTOMERS

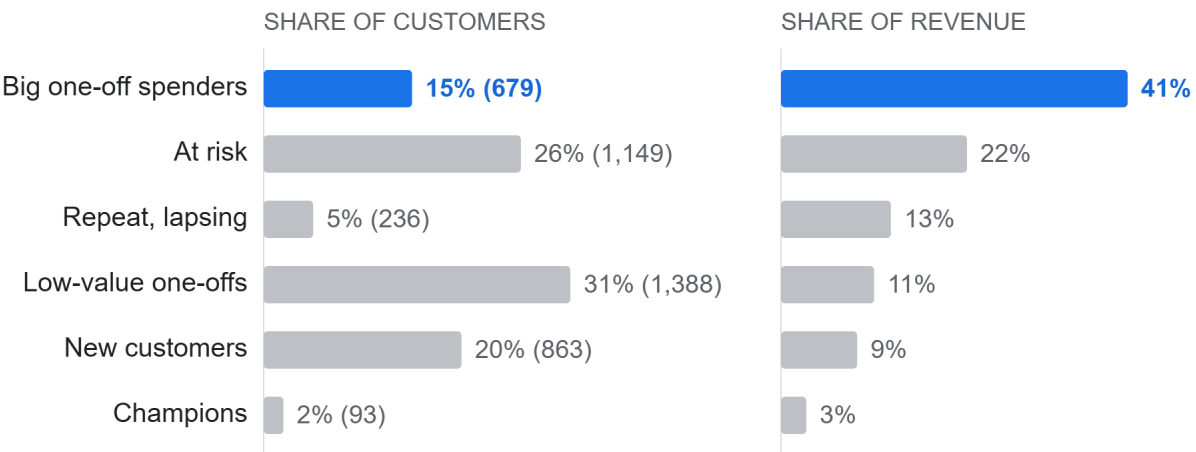
 5. Customers and retention

41%

of revenue comes from 15% of customers, each from a single shopping trip.

15% of customers bring in 41% of revenue

RFM segments: share of customers (with count) and share of revenue, 4,408 customers



Only **7.5%** of customers bought on more than one day in three months.

November's new customers came back more often than December's: **8.8% vs 5.7%** within 30 days.

→ **NEXT ACTION**

Most repeat buyers return within two weeks: send a second-purchase offer in that window.

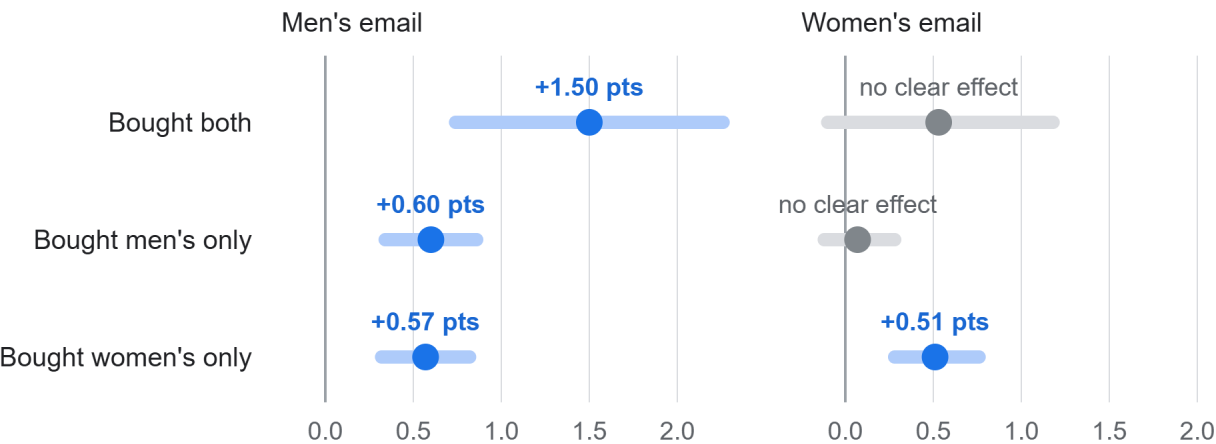
METHODS DEMONSTRATION EXTERNAL DATASET

▶ Email A/B test

64,000 customers randomly split: the men's email worked for everyone, the women's only for past women's-wear buyers.

The men's email moved everyone; the women's only past women's-wear buyers

Change in conversion vs no email, percentage points, 95% CI



The men's email took purchases from 0.57% to 1.25% and added \$0.77 per customer.

These numbers are about another retailer, not the Google store: GA4 has no experiments.

PLAN SIZES FIRST
A test this size only spots lifts of 39% or more.

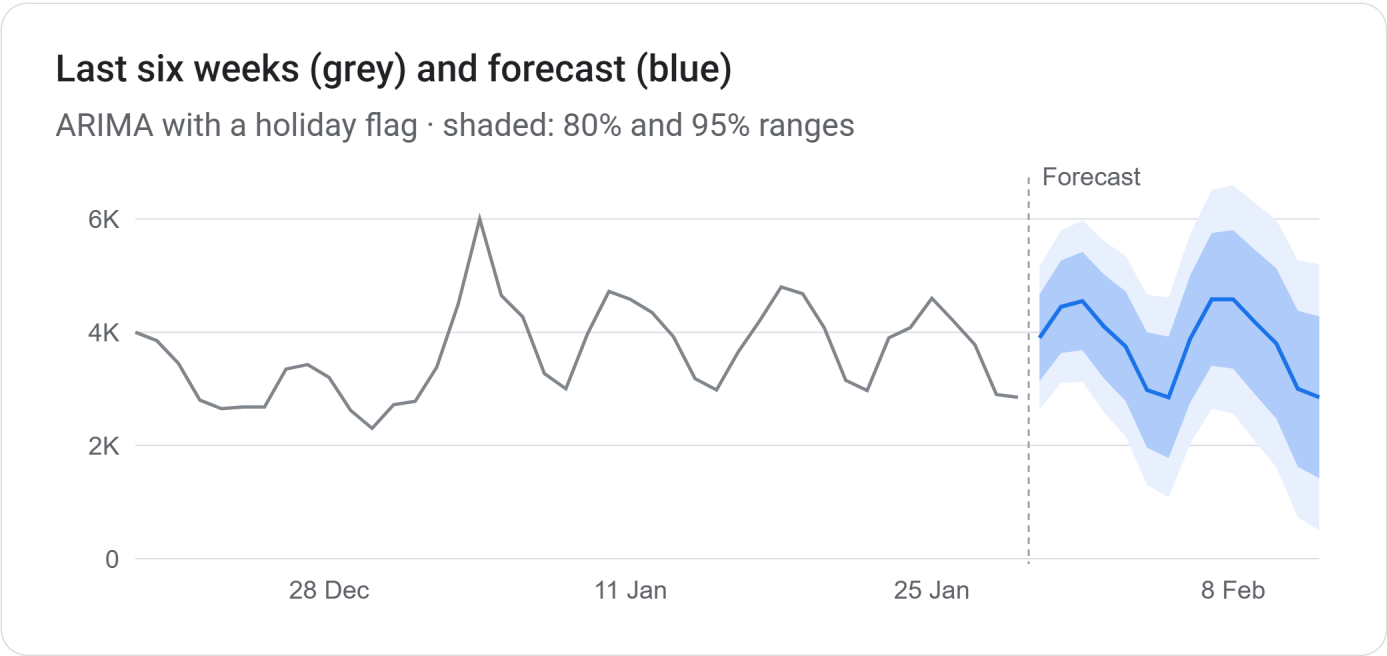
Source: Hillstrom email experiment (MineThatData, 2008), a different retailer. Holm-corrected

PLANNING

 Traffic forecast

3,800

sessions a day expected over the next two weeks.



The best model was off by about **18%** in two-week back-tests.

Daily orders can't be forecast from 92 days: every model was off by around 100%.

→ **NEXT ACTION**

Forecast visits, then apply a conversion assumption.

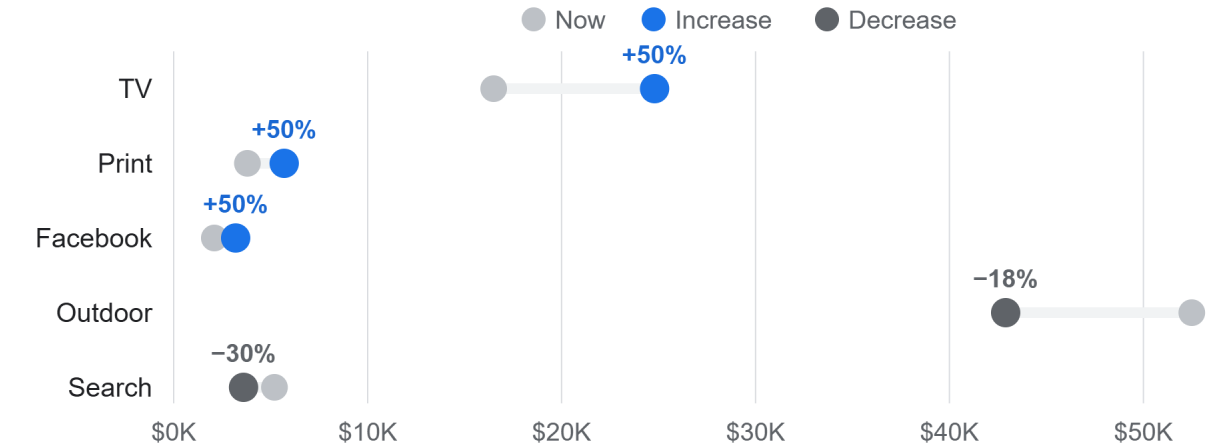
METHODS DEMONSTRATION EXTERNAL DATASET

Marketing mix model

+11% revenue from moving the same budget from outdoor and search to TV and print.

Same total budget, about 11% more revenue

Average weekly spend now and recommended · each channel kept within 70% to 150% of today



Outdoor takes 66% of spend and is saturated: an extra \$1 returns about 37 cents.

TV and print still return about \$2.70 and \$1.90 per extra \$1.

INTERPRETATION
The model explains 85% of revenue in weeks it never saw.

Source: Meta Robyn's simulated example data, 2016 to 2018. GA4 has no ad spend

WHAT TO DO NEXT



Five actions, in order of value for effort

#	ACTION	WHY	EFFORT
1	Fix GA4: exclude own domains, send transaction IDs, alert on broken tags	Every report on the raw data is wrong in some way	Low
2	Investigate the weak entry pages, then A/B test product pages; checkout second	3 in 4 product viewers never add to basket, on every device	Medium
3	Second-purchase offer within two weeks of a first order	Revenue relies on one-off buyers; repeats come back fast	Low
4	Protect SEO; holdout-test Paid Search before cutting it	Organic is about 40% of sales; Paid is undervalued by last click	Medium
5	Plan Q1 on a post-holiday baseline; investigate January conversion	Conversion fell 45% while traffic held up	Low

TECHNICAL APPENDIX



How it was built, and what the data can't tell us

Built with



Data: BigQuery to Parquet, modelled in DuckDB and dbt, with 62 automated data tests



Python: Attribution (Markov) and customer lifetime value, unit tested



R: A/B testing, forecasting and Meta's Robyn



Power BI: Five-page report generated from code, one design system across every output

Limits



Three months, with the holidays in the middle



Users are browsers, so cross-device journeys are split



About 9% of traffic sources are hidden by Google



No ad spend in GA4: the email test and mix model use separate datasets